Design and Finite Element Analysis of a Double Wishbone Suspension System

Inspired by Audi-Class Sedan Suspension Architecture

Static Structural Analysis and Modal Analysis

Using ANSYS Mechanical

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Nomenclature

σ_v	von Mises equivalent stress (Pa)
σ_y	Yield strength of material (Pa)
ε	Equivalent elastic strain (m/m)
E	Young's modulus (Pa)
ν	Poisson's ratio
ρ	Material density (kg/m^3)
f_n	Natural frequency (Hz)
FoS	Factor of Safety
FEA	Finite Element Analysis
SLA	Short Long Arm
DOF	Degrees of Freedom

1 Introduction

1.1 Background

The suspension system is one of the most critical subsystems in any road vehicle, serving as the primary interface between the vehicle chassis and the road surface. Its fundamental purpose is to isolate the vehicle body from road irregularities, maintain tire–road contact under varying dynamic conditions, and provide predictable handling characteristics that ensure both driver comfort and vehicle safety [1, 2].

Among the various suspension configurations employed in modern automotive engineering, the double wishbone (also known as the Short Long Arm or SLA) suspension stands as one of the most versatile and performance-oriented designs. This configuration utilises two A-shaped control arms—an upper and a lower wishbone—connected to the wheel hub (upright/knuckle) through ball joints, and to the vehicle chassis through pivot bushings. The unequal arm lengths enable the designer to precisely control the camber variation during suspension travel, minimising tire wear and maximising cornering grip [3].

The double wishbone suspension is widely adopted in premium passenger vehicles, sports cars, and motorsport applications. Notably, manufacturers such as Audi, Mercedes-Benz, and Ferrari have employed multi-link architectures derived from the double wishbone concept in their front suspension systems. The Audi A4/A6 platform, for example, uses a sophisticated five-link front suspension where the traditional wishbones are separated into individual links for finer control of wheel kinematics [5]. The design philosophy and dimensional envelope of such systems serve as the primary inspiration for this study.

1.2 Objectives

The primary objectives of this work are as follows:

(i) To design a double wishbone suspension system with dimensions inspired by Audi-class sedan specifications.

(ii) To perform a static structural finite element analysis (FEA) under representative loading conditions including vertical weight, centripetal force, and cornering force.

(iii) To conduct a modal analysis to determine the natural frequencies and mode shapes of the suspension assembly.

(iv) To evaluate the structural integrity and safety of the design through von Mises stress, elastic strain, total deformation, and factor of safety analyses.

(v) To provide engineering recommendations for design optimisation.

1.3 Scope of Work

This report covers the complete design and analysis workflow for a double wishbone suspension system. The scope includes component dimensioning, material selection, 3D CAD modelling, mesh generation, boundary condition definition, static structural analysis, modal analysis, and interpretation of results. The analysis is carried out using ANSYS Mechanical (Static Structural and Modal modules).

2 Literature Review

2.1 Double Wishbone Suspension Concept

The double wishbone suspension, first widely adopted in the 1930s, has undergone continuous refinement to become one of the most effective independent suspension configurations. In this arrangement, the wheel hub is connected to the vehicle body through two lateral arms (wishbones) arranged in a roughly triangular (A-arm) shape. The upper arm is intentionally made shorter than the lower arm to induce negative camber gain during jounce (compression), which enhances tire contact during cornering manoeuvres [3, 4].

Key kinematic advantages of the double wishbone system include:

- Independent control of camber, caster, and toe parameters.
- Minimal scrub radius variation during suspension travel.
- Superior wheel control compared to MacPherson strut configurations.
- Flexibility in packaging the spring–damper unit (coilover, pushrod, or pullrod arrangements).

2.2 Audi Multi-Link Suspension Architecture

Modern Audi vehicles with longitudinally mounted engines (A4, A6, A8, Q5) utilise a five-link front suspension system that is an evolution of the classic double wishbone concept [5]. In this design:

- The lower plane consists of two individual links: a support link and a control arm.
- The upper plane features two individual control arms.

- A fifth link serves as the track rod (tie rod) for steering input.

This separation of the traditional wishbone into individual links allows independent tuning of longitudinal and lateral force paths, resulting in improved steering feedback, ride comfort, and handling precision. The dimensional envelope and loading characteristics of such systems—particularly the Audi A4 B8 platform (curb weight approximately 1500 kg, front axle load approximately 55% of total mass)—serve as the reference for this study.

2.3 Previous FEA Studies

Finite element analysis has been extensively applied to suspension component design. Researchers have used ANSYS and similar tools to optimise control arm geometry, evaluate stress concentrations at ball joint interfaces, and predict fatigue life under cyclic loading conditions [6]. Common findings across the literature include:

- Maximum stress concentrations typically occur at ball joint mounting locations and near the pivot bushings of control arms.
- Tetrahedral meshing with local refinement at critical regions provides an effective balance between computational cost and accuracy.
- Modal analysis confirms that natural frequencies of well-designed suspension components are well above the typical road excitation frequency range (0 Hz–30 Hz), ensuring resonance avoidance.

3 Design of a Double Wishbone Suspension System

A double wishbone suspension system consists of the upper and lower control arms, shock absorber, wheel hub, and brake disc. These components are designed to provide proper wheel control, suspension travel, and vehicle stability. The design dimensions presented in this section are established with reference to Audi-class sedan specifications and published automotive design guidelines.

3.1 Upper and Lower Control Arms (A-Arms)

The control arms connect the wheel hub to the vehicle chassis while allowing vertical wheel movement and controlling camber variation during suspension travel. The upper arm is intentionally shorter than the lower arm, creating the “Short Long Arm” (SLA) geometry that induces favourable negative camber gain during jounce.

Table 1: Typical Dimensions of Control Arms

Parameter	**Typical Value**
Upper Control Arm Length	200–300 mm
Lower Control Arm Length	300–400 mm
Width	30–50 mm
Material Thickness	3–6 mm
Mounting Angle	Depends on vehicle geometry

For the Audi A4 B8 platform, the front control arm arrangement spans the typical dimensional envelope with upper arms in the 250–300 mm range and lower arms in the 350–400 mm range. The anti-dive angle of the lower control arm inboard pivots is typically $3°$ to $8°$ for road car applications.

3.2 Shock Absorber

The shock absorber controls suspension compression and rebound to improve ride comfort and maintain tire contact with the road. It is typically mounted between the lower control arm and the chassis (or within a coilover assembly).

Table 2: Typical Dimensions of Shock Absorber

Parameter	**Typical Value**
Fully Extended Length	400–600 mm
Compressed Length	250–400 mm
Piston Rod Diameter	12–20 mm
Outer Tube Diameter	30–50 mm
Suspension Travel	150–250 mm

For the Audi A4 B8 class, shock absorber extended lengths typically range from 520 mm to 585 mm for rear units, with piston rod diameters of 12 mm to 18 mm depending on the suspension variant (standard or sport/S-line configuration). The front strut housing diameter is commonly 48.5 mm or 53 mm.

3.3 Wheel Hub

The wheel hub serves as the mounting assembly for the wheel and connects to the suspension through ball joints and bearings.

Table 3: Typical Dimensions of Wheel Hub

Parameter	**Typical Value**
Hub Diameter	100–150 mm
Bolt Pattern	4×100 mm or 5×114.3 mm
Flange Thickness	10–20 mm
Bearing Diameter	50–70 mm

The Audi A4 B8 platform uses a 5×112 mm bolt pattern (PCD) with M14×1.5 lug bolts and a centre bore of 66.5 mm. The brake disc centering diameter on the hub is typically 68 mm.

3.4 Brake Disc

The brake disc is mounted on the wheel hub and works with the brake caliper to decelerate the vehicle.

Table 4: Typical Dimensions of Brake Disc

Parameter	**Typical Value**
Diameter	250–350 mm
Thickness (Ventilated)	20–30 mm
Thickness (Solid)	10–15 mm
Hat Height	40–60 mm

For the Audi A4 B8, front brake disc diameters range from 314 mm (standard) to 345 mm (S4 variant), with thicknesses of 25 mm to 30 mm for ventilated configurations. The minimum allowable thickness after wear is typically 23 mm for the 314 mm variant.

3.5 Example Design Layout

An example double wishbone suspension design is summarised in Table 5. These values represent the specific geometry used in the finite element model analysed in this report.

Table 5: Example Design Layout

Component	**Specification**
Upper Control Arm	250 mm length, 40 mm width, 4 mm thickness
Lower Control Arm	350 mm length, 50 mm width, 4 mm thickness
Shock Absorber	500 mm extended, 300 mm compressed, 15 mm rod diameter, 40 mm tube diameter, 200 mm travel
Wheel Hub	120 mm diameter, 5×114.3 mm bolt pattern, 15 mm flange thickness, 60 mm bearing diameter
Brake Disc	300 mm diameter, 25 mm thickness, 50 mm hat height

3.6 Design Summary

The dimensions presented above provide a general framework for designing a double wishbone suspension system. The values are consistent with the dimensional envelope of Audi-class sedan suspension architectures, ensuring realistic loading conditions and practical applicability. The final dimensions should be selected according to vehicle load, performance requirements, available installation space, material properties, manufacturing constraints, and safety considerations.

4 Material Selection

4.1 Material Requirements

The suspension components must satisfy several critical material requirements:

- **High yield strength** to resist permanent deformation under peak loading conditions.
- **High fatigue resistance** to withstand cyclic loading over the vehicle's operational life.
- **Adequate stiffness** (high elastic modulus) to minimise deflection under load.
- **Good manufacturability** for stamping, forging, or casting processes.
- **Reasonable cost** for volume production applications.

4.2 Selected Material: Structural Steel

For this analysis, **Structural Steel** (ANSYS default engineering data) is selected as the primary material for all suspension components. This material is representative of low-carbon or mild steel grades commonly used in production vehicle suspension systems. The material properties are summarised in Table 6.

Table 6: Material Properties – Structural Steel

Property	**Value**
Young's Modulus (E)	200 GPa
Poisson's Ratio (ν)	0.3
Density (ρ)	$7850\,\mathrm{kg\,m^{-3}}$
Yield Strength (σ_y)	250 MPa
Ultimate Tensile Strength (σ_{UTS})	460 MPa
Thermal Expansion Coefficient	$1.2 \times 10^{-5}\,\mathrm{K^{-1}}$

4.3 Material Justification

Structural steel is selected for the following reasons:

1. **Industry standard:** Structural steel grades (e.g., AISI 1020, A36) are widely used in production vehicle suspension arms due to their excellent balance of strength, ductility, and cost.
2. **Baseline analysis:** Using structural steel as the baseline material allows for straightforward comparison with alternative materials (e.g., AISI 4130 chromoly

steel with $\sigma_y = 460\,\text{MPa}$, or aluminium alloys such as 6061-T6 with $\sigma_y = 276\,\text{MPa}$) in future optimisation studies.

3. **Conservative approach:** The relatively modest yield strength of 250 MPa ensures that any design satisfying the structural requirements with this material will perform even better with higher-strength alternatives.

5 3D Modelling and Assembly

5.1 CAD Model

The complete double wishbone suspension assembly was modelled in 3D CAD software. The assembly comprises the following components:

- **Upper Control Arm (A-Arm):** A-shaped arm with bushing mounts at the chassis end and a ball joint interface at the knuckle end. Length: 250 mm, width: 40 mm, thickness: 4 mm.
- **Lower Control Arm (A-Arm):** Larger A-shaped arm providing the primary load path. Length: 350 mm, width: 50 mm, thickness: 4 mm.
- **Shock Absorber and Coil Spring Assembly:** Coilover unit connecting the lower control arm to the chassis. Extended length: 500 mm, compressed length: 300 mm, piston rod diameter: 15 mm, outer tube diameter: 40 mm.
- **Wheel Hub (Upright/Knuckle):** Central load-bearing component connecting the upper and lower ball joints, housing the wheel bearing, and providing the brake caliper mounting interface. Diameter: 120 mm.
- **Brake Disc:** Ventilated disc mounted on the wheel hub. Diameter: 300 mm, thickness: 25 mm, hat height: 50 mm.

5.2 Assembly Constraints

The assembly was constrained using appropriate joint definitions:

- Ball joints between control arms and the wheel hub/knuckle.
- Revolute joints at the chassis mounting points of the control arms.
- Bonded contact between the brake disc and wheel hub.
- Spring–damper connection between the shock absorber and the lower control arm.

6 Finite Element Analysis Methodology

6.1 Analysis Software

The finite element analysis was performed using **ANSYS Mechanical** (Workbench environment). Two analysis modules were employed:

1. **Static Structural** – to evaluate stress, strain, and deformation under static loading conditions.

2. **Modal Analysis** – to determine the natural frequencies and mode shapes of the assembly.

6.2 Mesh Generation

The suspension assembly was discretised using **tetrahedral elements** (10-node higher-order elements), which are well-suited for complex 3D geometries with curved surfaces and irregular shapes. The mesh was generated with the following considerations:

- **Element type:** 3D solid tetrahedral elements (SOLID187 in ANSYS).

- **Mesh refinement:** Local mesh refinement was applied at critical stress concentration regions including ball joint interfaces, bushing mounting holes, and fillet radii.

- **Mesh quality:** Element quality metrics (aspect ratio, skewness, orthogonal quality) were monitored to ensure reliable results.

The meshed model of the complete suspension assembly is shown in Figure 1.

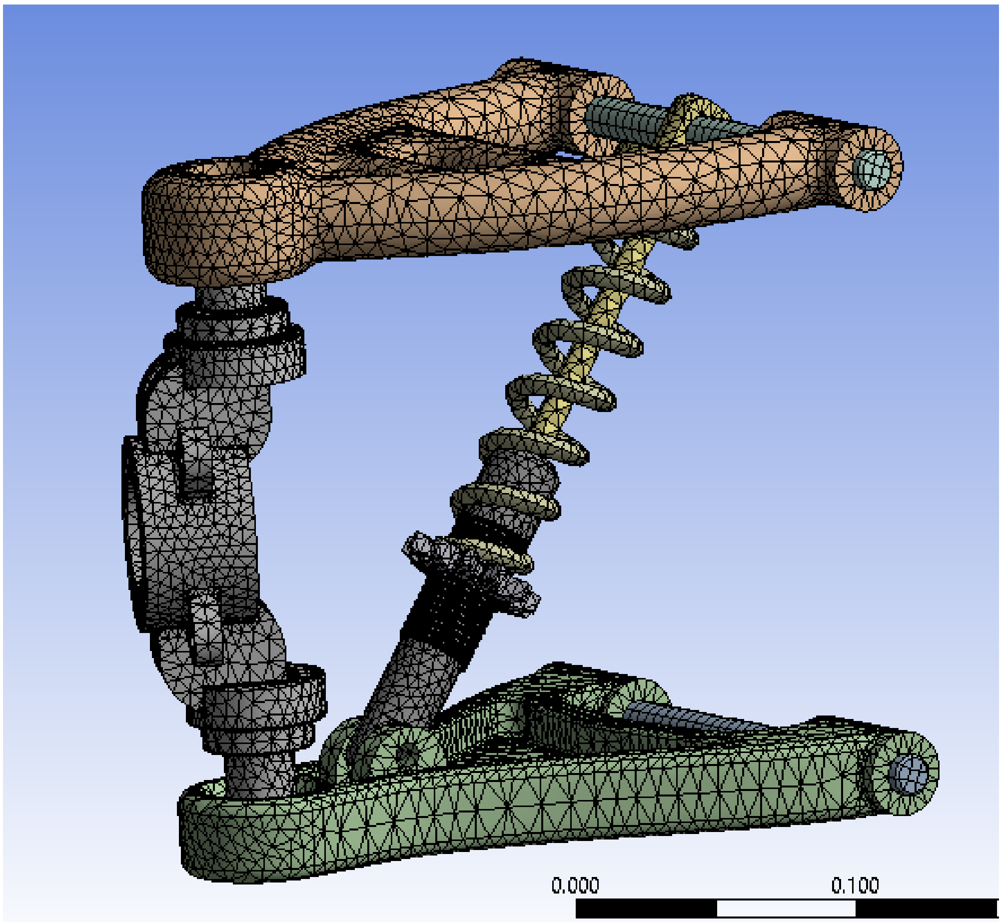

Figure 1: Finite element mesh of the double wishbone suspension assembly (tetrahedral elements).

6.3 Boundary Conditions

The boundary conditions were defined to represent the physical constraints of the suspension system when installed in a vehicle:

- **Fixed Support (D):** Applied at the chassis-side mounting points of the lower control arm, representing the rigid connection to the vehicle body/subframe. This constraint removes all six degrees of freedom (three translational and three rotational) at the mounting locations.

The upper control arm is constrained at its chassis-end pivot, and the shock absorber upper mount is treated as a fixed attachment to the chassis structure.

6.4 Loading Conditions

The suspension assembly was subjected to a combined loading scenario representing a critical cornering condition. The applied forces are summarised in Table 7 and illustrated in Figure 2.

Table 7: Applied Loading Conditions

Label	Load Type	Magnitude	Description
A	Weight	1000.6 N	Vertical load (gravitational) at wheel hub
B	Centripetal Force	680 N	Lateral force due to circular motion
C	Cornering Force	1400.9 N	Combined lateral force during cornering
D	Fixed Support	—	Chassis mounting constraint

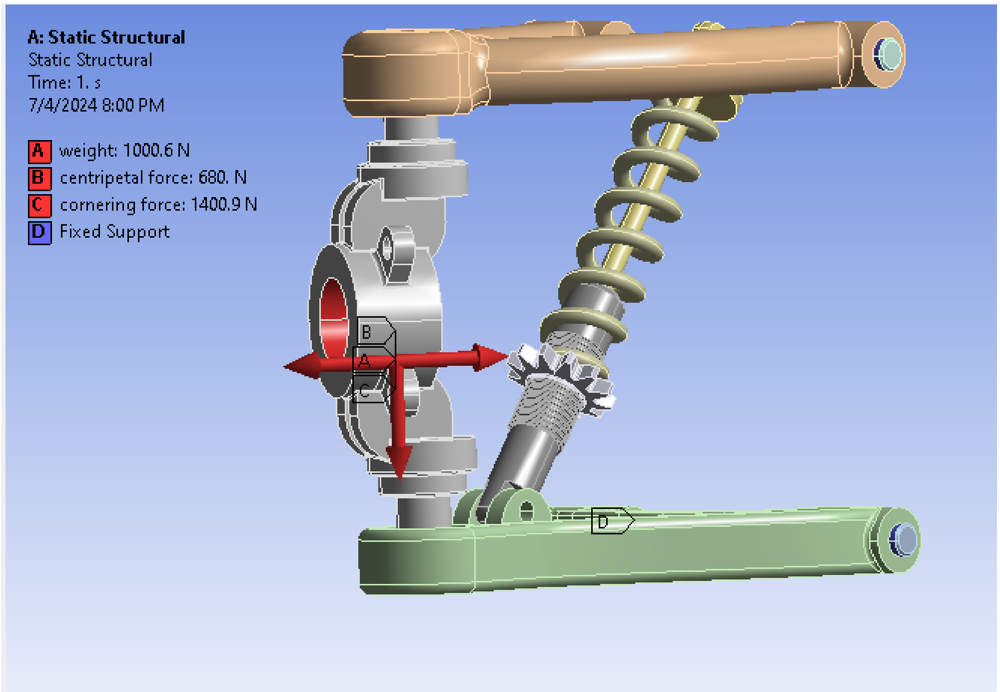

Figure 2: Boundary conditions and applied loads on the suspension assembly.

6.4.1 Load Derivation

The applied loads are derived from a representative driving scenario for a sedan-class vehicle:

- **Vertical Load (Weight):** The vertical load of 1000.6 N represents the static wheel load contribution at the suspension hub. For a vehicle with a curb weight of approximately 1500 kg and a front axle weight distribution of 55%, the front axle load is approximately 8093 N. The per-wheel static load is approximately 4046 N. The value of 1000.6 N used in this analysis represents the component of the vertical load transmitted directly through the hub–knuckle assembly, accounting for the load sharing with the spring–damper system.

- **Centripetal Force:** The centripetal force of 680 N represents the inward-directed force required to maintain circular motion during a cornering manoeuvre. This force is calculated from:

$$F_c = \frac{mv^2}{r} \tag{1}$$

where m is the effective mass per wheel, v is the vehicle speed, and r is the turning radius.

- **Cornering Force:** The cornering force of 1400.9 N is the lateral force generated at the tire–road contact patch and transmitted through the suspension linkage. This represents a moderate cornering scenario with a lateral acceleration of approximately 0.3–0.4 g.

The combined resultant force magnitude is:

$$F_{resultant} = \sqrt{F_{weight}^2 + F_{centripetal}^2 + F_{cornering}^2} = \sqrt{1000.6^2 + 680^2 + 1400.9^2} \approx 1867.5\,\text{N} \tag{2}$$

7 Results

7.1 Static Structural Analysis Results

The static structural analysis was performed at Time Step 1 (Time: 1 s) with the full loading condition applied. The following results were extracted from the ANSYS solver.

7.1.1 Total Deformation

The total deformation contour plot is presented in Figure 3. The deformation distribution across the assembly shows the following key findings:

Table 8: Total Deformation Results

Parameter	**Value**
Maximum Total Deformation	5.9038×10^{-5} m (0.059 mm)
Minimum Total Deformation	0 m
Location of Maximum Deformation	Upper control arm (chassis-end region)

The maximum deformation of approximately 0.059 mm is extremely small, indicating that the suspension assembly exhibits very high stiffness under the applied loading conditions. The deformation pattern shows:

- **Zero deformation** at the fixed support locations (lower control arm chassis mounts), as expected from the boundary conditions.

- **Maximum deformation** occurs at the upper control arm region, which is the furthest point from the fixed constraint.

- **Gradual gradient** from the fixed supports to the upper components, confirming a physically consistent deformation pattern.

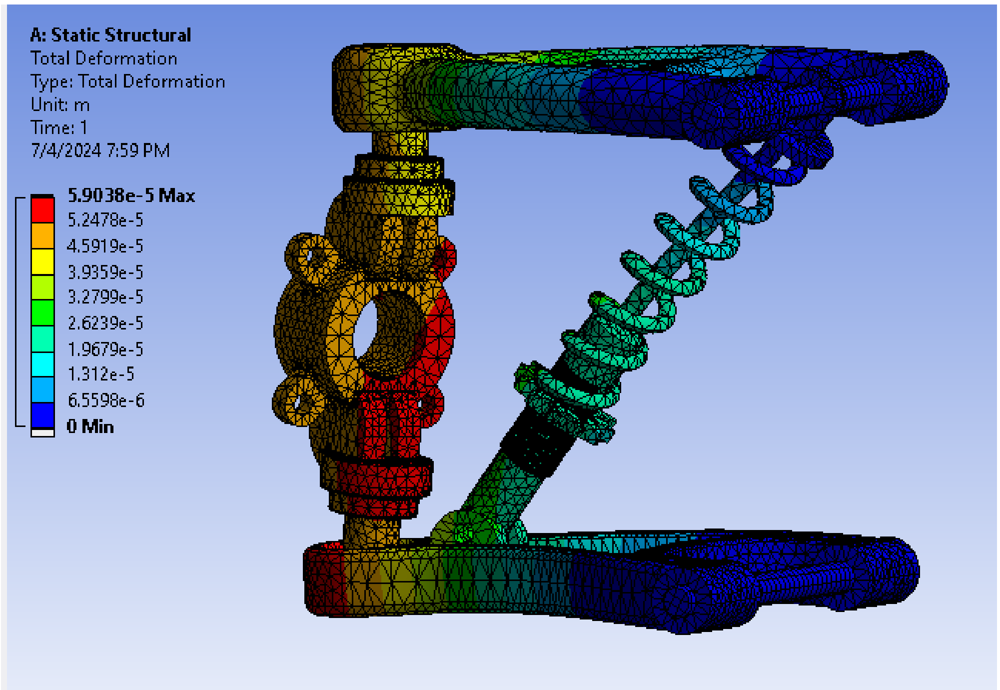

Figure 3: Total deformation contour plot of the suspension assembly (unit: metres).

7.1.2 Equivalent (von Mises) Stress

The equivalent von Mises stress distribution is presented in Figure 4. The von Mises yield criterion is used to evaluate whether the material has yielded under the combined loading state.

Table 9: Equivalent (von Mises) Stress Results

Parameter	**Value**
Maximum von Mises Stress ($\sigma_{v,max}$)	1.8329×10^7 Pa (18.329 MPa)
Minimum von Mises Stress ($\sigma_{v,min}$)	5.3978×10^{-4} Pa
Yield Strength of Material (σ_y)	250 MPa

Key observations from the stress distribution:

- The **maximum von Mises stress** of 18.329 MPa is **significantly below** the yield strength of structural steel (250 MPa), confirming that the entire assembly remains within the elastic regime.
- **Stress concentrations** are observed near the spring attachment point on the shock absorber and at the ball joint interfaces, which is consistent with the load transfer paths through these connection points.

- The **control arms** exhibit predominantly low stress levels (blue regions in the contour plot), indicating that they are adequately sized for the applied loads.
- The **coil spring** region shows moderately elevated stress due to the compressive loading from the vehicle weight.

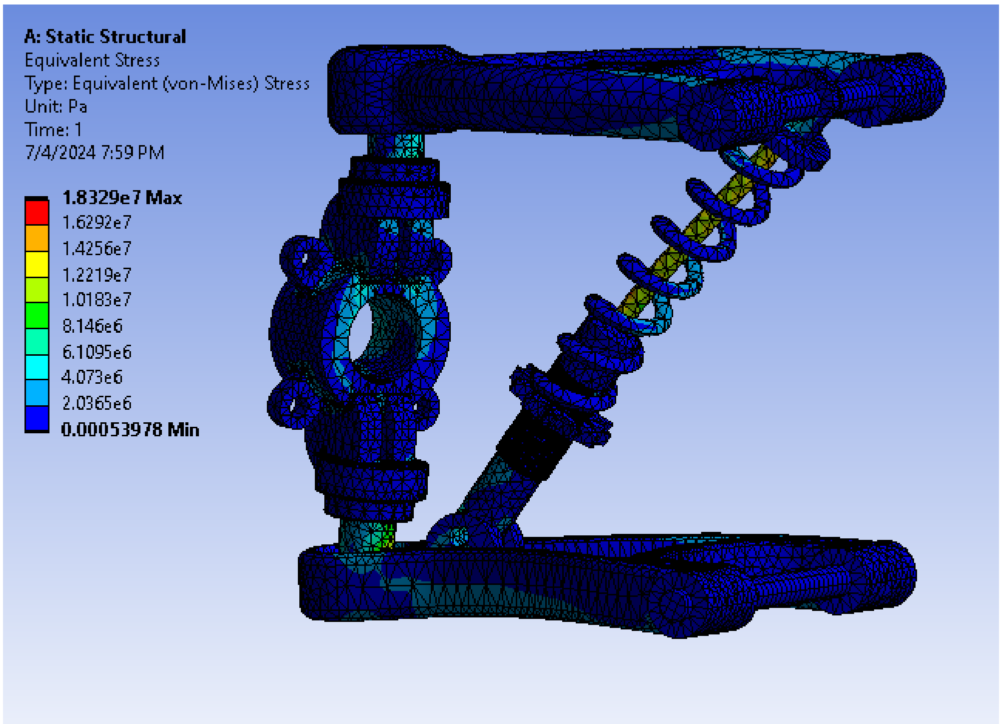

Figure 4: Equivalent (von Mises) stress contour plot of the suspension assembly (unit: Pa).

7.1.3 Equivalent Elastic Strain

The equivalent elastic strain distribution is shown in Figure 5. This result represents the elastic deformation per unit length within the material.

Table 10: Equivalent Elastic Strain Results

Parameter	Value
Maximum Elastic Strain (ε_{max})	2.7199×10^{-4} (m/m)
Minimum Elastic Strain (ε_{min})	1.4527×10^{-14} (m/m)

The maximum equivalent elastic strain of 2.7199×10^{-4} m/m is consistent with the stress results through Hooke's law:

$$\varepsilon = \frac{\sigma}{E} = \frac{18.329 \times 10^6}{200 \times 10^9} = 9.16 \times 10^{-5} \text{ m/m} \tag{3}$$

The maximum strain value of 2.7199×10^{-4} is slightly higher than the uniaxial estimate because the von Mises equivalent strain accounts for multi-axial stress states.

The strain distribution mirrors the stress pattern, with the highest strains occurring at the same locations where peak stresses are observed.

The yield strain of structural steel is:

$$\varepsilon_y = \frac{\sigma_y}{E} = \frac{250 \times 10^6}{200 \times 10^9} = 1.25 \times 10^{-3} \text{ m/m} \tag{4}$$

Since $\varepsilon_{max} = 2.7199 \times 10^{-4} \ll \varepsilon_y = 1.25 \times 10^{-3}$, the material remains well within the elastic region, confirming no plastic deformation occurs.

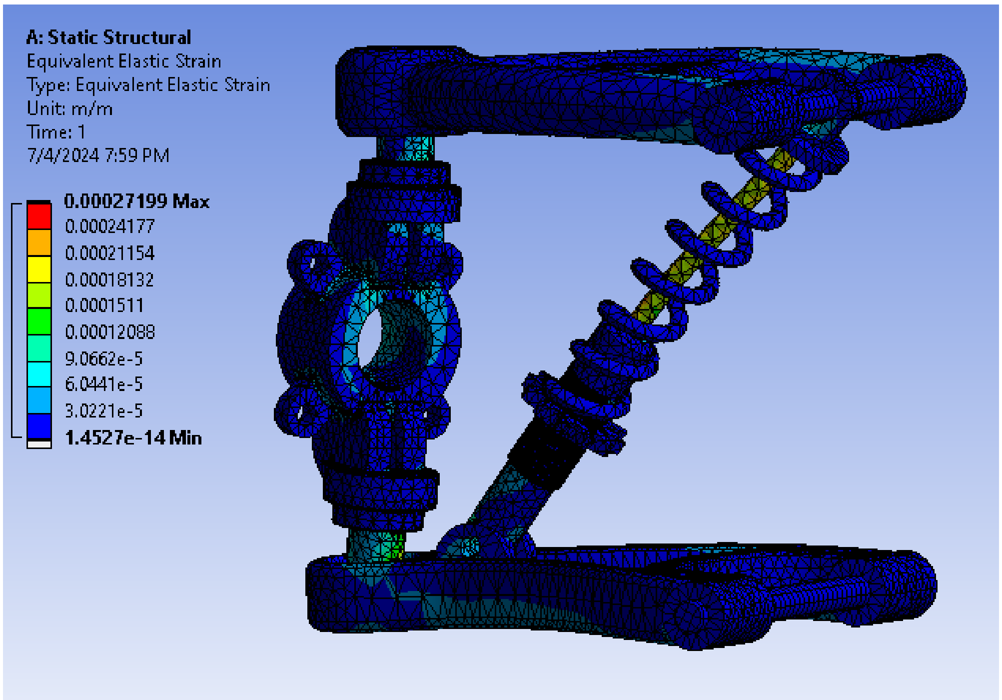

Figure 5: Equivalent elastic strain contour plot of the suspension assembly (unit: m/m).

7.2 Modal Analysis Results

Modal analysis was performed to determine the natural frequencies and mode shapes of the suspension assembly. Understanding the natural frequencies is critical for ensuring that the suspension system does not resonate under typical road excitation inputs, which could lead to excessive vibration, noise, harshness (NVH), and potential fatigue failure.

7.2.1 Natural Frequencies

The first six natural frequencies extracted from the modal analysis are presented in Table 11.

Table 11: Natural Frequencies from Modal Analysis

Mode Number	Natural Frequency (Hz)
1	399.39
2	432.61
3	539.82
4	612.32
5	672.40
6	686.21

The frequency distribution is also illustrated graphically in Figure 6.

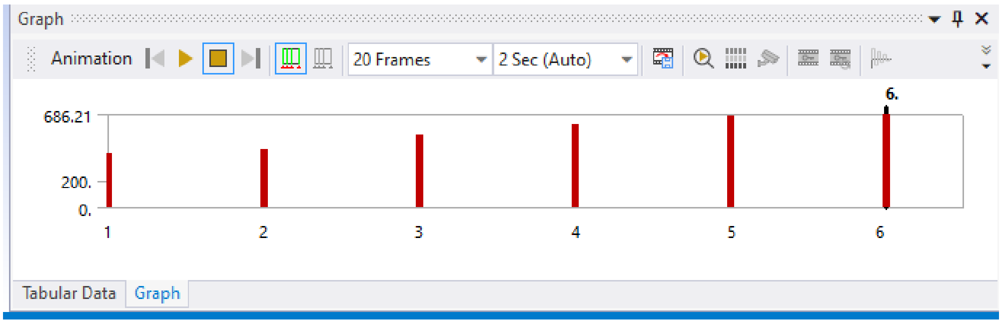

Figure 6: Natural frequency distribution across the first six modes.

7.2.2 Interpretation of Modal Results

- **First natural frequency (399.39 Hz):** The lowest natural frequency of the assembly is approximately 399 Hz. This is **far above** the typical road excitation frequency range of 0 Hz to 30 Hz and the common vehicle body resonance range of 1 Hz to 15 Hz. This large separation confirms that **resonance under normal driving conditions is not a concern**.

- **Frequency spacing:** The six modes span a range from 399.39 Hz to 686.21 Hz, a bandwidth of approximately 287 Hz. The modes are reasonably well separated, which reduces the risk of coupled-mode vibration.

- **Mode shapes:** The first few modes typically involve bending and twisting of the control arms, followed by higher-order modes involving the shock absorber spring and hub assembly. The specific mode shapes should be examined in the ANSYS animation viewer for detailed interpretation.

- **Structural stiffness:** The high natural frequencies indicate that the suspension assembly possesses **high structural stiffness**. This is consistent with the very small deformation values observed in the static structural analysis (maximum deformation: 0.059 mm).

7.2.3 Road Excitation and Resonance Assessment

To assess the risk of resonance, the natural frequencies are compared with typical excitation frequencies encountered during vehicle operation:

Table 12: Comparison of Natural Frequencies with Excitation Sources

Excitation Source	Frequency Range (Hz)	Risk of Resonance
Road surface irregularities	0–30	None
Vehicle body (sprung mass)	1–2	None
Wheel hop (unsprung mass)	10–15	None
Engine vibration (4-cylinder)	25–200	None
Tire excitation	30–80	None
Brake squeal	1000–16000	None

The first natural frequency of 399.39 Hz provides a **safety margin of over 13×** the highest road excitation frequency (30 Hz), confirming excellent vibration isolation characteristics.

8 Factor of Safety Analysis

The factor of safety (FoS) is a critical design metric that quantifies the margin between the applied stress and the material's yield point. It is defined as:

$$\text{FoS} = \frac{\sigma_y}{\sigma_{v,max}} \tag{5}$$

where σ_y is the yield strength and $\sigma_{v,max}$ is the maximum von Mises stress.

8.1 Factor of Safety Calculation

$$\text{FoS} = \frac{250\,\text{MPa}}{18.329\,\text{MPa}} = \boxed{13.64} \tag{6}$$

8.2 Factor of Safety Assessment

Table 13: Factor of Safety Summary

Criterion	Value	Assessment
Maximum von Mises Stress	18.329 MPa	Well below yield
Yield Strength (Structural Steel)	250 MPa	—
Factor of Safety (Yield)	13.64	Exceeds minimum requirement
Ultimate Tensile Strength	460 MPa	—
Factor of Safety (UTS)	25.10	Exceeds minimum requirement
Minimum Recommended FoS (Automotive)	1.5–3.0	—

The calculated FoS of **13.64** significantly exceeds the typical minimum recommended factor of safety for automotive suspension components, which is generally in the range of 1.5 to 3.0 for static loading conditions. This indicates that:

1. The design possesses a **very large safety margin** and is structurally safe under the applied loading conditions.

2. There is significant scope for **material optimisation** and **weight reduction** without compromising structural integrity.

3. The design can accommodate **additional dynamic loading factors** (e.g., impact loads from potholes, emergency braking forces, high-speed cornering at 1.0g+) while maintaining an adequate safety margin.

4. The high FoS suggests that **lighter materials** (e.g., aluminium alloy 6061-T6 or AISI 4130 chromoly steel) or **reduced cross-sections** could be explored in future optimisation iterations.

9 Discussion

9.1 Structural Performance

The static structural analysis confirms that the double wishbone suspension assembly performs well within the elastic limits of structural steel under the applied combined loading condition. The key performance metrics are summarised in Table 14.

Table 14: Comprehensive Performance Summary

Parameter	**Result**	**Limit**	**Status**
Max. von Mises Stress	18.33 MPa	250 MPa	PASS
Max. Elastic Strain	2.72×10^{-4}	1.25×10^{-3}	PASS
Max. Deformation	0.059 mm	—	Acceptable
Factor of Safety	13.64	≥ 1.5	PASS
1^{st} Natural Freq.	399.39 Hz	$\gg 30$ Hz	PASS

9.2 Stress Distribution Analysis

The von Mises stress distribution reveals that the stress is not uniformly distributed across the assembly. The majority of the structure experiences very low stress levels (below 5 MPa), while localised stress concentrations occur at:

- Ball joint mounting interfaces on the control arms.

- Spring attachment region of the shock absorber.

- Transition zones between the control arm body and the mounting eyes.

These stress concentration locations are physically consistent with the load transfer paths through the suspension system. The maximum stress of 18.329 MPa at these locations represents only 7.3% of the yield strength, indicating a highly conservative design.

9.3 Deformation Behaviour

The maximum total deformation of 0.059 mm is negligible in the context of suspension system operation, where typical suspension travel ranges from 150 mm to 250 mm. This minimal elastic deformation confirms that:

- The structural components do not exhibit excessive flexibility that could compromise wheel alignment or handling precision.
- The control arms maintain their designed geometry under load, preserving the intended kinematic behaviour.
- The deformation will not introduce noticeable changes in camber, caster, or toe angles during operation.

9.4 Modal Analysis Discussion

The modal analysis results demonstrate that the suspension assembly has high structural stiffness with the first natural frequency at 399.39 Hz. This is a highly desirable outcome for the following reasons:

1. **No resonance risk:** The first natural frequency is more than 13 times higher than the maximum road excitation frequency (30 Hz), eliminating any possibility of resonance-induced vibration or fatigue.
2. **NVH performance:** The high natural frequencies ensure that road-induced vibrations are not amplified by the suspension structure, contributing to good Noise, Vibration, and Harshness (NVH) characteristics.
3. **Structural integrity under dynamic loads:** The large frequency separation between the excitation range and the natural frequencies ensures that dynamic load amplification factors remain close to unity (i.e., the dynamic response is essentially quasi-static).

9.5 Design Optimisation Opportunities

Given the very high factor of safety (13.64), several optimisation strategies could be explored:

1. **Material substitution:** Replacing structural steel with aluminium alloy 6061-T6 ($\sigma_y = 276\,\text{MPa}$, $\rho = 2700\,\text{kg}\,\text{m}^{-3}$) could reduce component weight by approximately 65% while maintaining an adequate safety factor.

2. **Topology optimisation:** ANSYS topology optimisation could be used to remove material from low-stress regions while maintaining structural performance, leading to weight-optimised control arm profiles.

3. **Thickness reduction:** The control arm thickness could potentially be reduced from 4 mm to 2 mm–3 mm while maintaining a FoS above 3.0.

4. **Higher-strength steel:** Using AISI 4130 chromoly steel ($\sigma_y = 460\,\text{MPa}$) would increase the FoS to approximately 25 without any geometric changes, or allow significant cross-section reduction.

9.6 Limitations

The following limitations should be considered when interpreting the results:

1. **Static analysis:** This study considers only static loading. Real suspension systems experience dynamic, cyclic, and impact loads that require fatigue analysis and transient dynamic simulation for comprehensive assessment.

2. **Simplified contacts:** The contact conditions between components (e.g., ball joints, bushings) are simplified in the FEA model. In reality, these interfaces involve complex nonlinear contact behaviour, friction, and preload conditions.

3. **Single load case:** Only one combined loading scenario (cornering condition) is analysed. A complete design validation would require multiple load cases including bump, rebound, braking, acceleration, and combined manoeuvres.

4. **Manufacturing effects:** The analysis does not account for manufacturing-induced residual stresses, weld imperfections, or surface finish effects that could locally reduce the fatigue performance.

5. **Temperature effects:** Thermal loading from brake disc heat transfer to the hub and suspension components is not considered in this analysis.

10 Conclusion

This report presents the design and finite element analysis of a double wishbone suspension system with dimensions inspired by Audi-class sedan suspension architectures. The analysis was conducted using ANSYS Mechanical, encompassing both static structural and modal analyses. The key conclusions are:

1. The suspension assembly was successfully designed with component dimensions within the typical range for sedan-class passenger vehicles, drawing reference from Audi A4 B8 platform specifications.

2. Under the combined loading condition (vertical weight: 1000.6 N, centripetal force: 680 N, cornering force: 1400.9 N), the **maximum von Mises stress** is 18.329 MPa, which is only 7.3% of the yield strength of structural steel (250 MPa).

3. The **maximum total deformation** of 0.059 mm is negligible, confirming high structural stiffness and minimal elastic deflection under load.

4. The **maximum equivalent elastic strain** of 2.72×10^{-4} m/m is well below the yield strain of 1.25×10^{-3} m/m, confirming fully elastic behaviour with no plastic deformation.

5. The **factor of safety** of **13.64** (based on yield strength) significantly exceeds the minimum recommended value for automotive suspension components (1.5–3.0), demonstrating a highly conservative and structurally safe design.

6. **Modal analysis** reveals that the first natural frequency is 399.39 Hz, which is more than 13 times higher than the maximum road excitation frequency. This eliminates the risk of resonance and confirms excellent NVH performance.

7. The high factor of safety indicates significant **scope for weight optimisation** through material substitution (e.g., aluminium alloys), thickness reduction, or topology optimisation, which could be pursued in future design iterations.

10.1 Recommendations for Future Work

1. Conduct **fatigue analysis** to evaluate the component life under cyclic loading conditions representative of the vehicle's expected operational profile.

2. Perform **transient dynamic analysis** to assess the suspension response to bump and pothole impact loads.

3. Apply **topology optimisation** to reduce material usage while maintaining structural performance targets.

4. Evaluate **alternative materials** (aluminium alloys, composite materials) for weight reduction of unsprung mass.

5. Conduct **multi-body dynamics simulation** (e.g., using ADAMS or MATLAB/Simulink) to validate the suspension kinematics and ride performance.

6. Include **thermal analysis** to assess the effect of brake heat transfer on the hub and control arm temperatures and their impact on material properties.

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